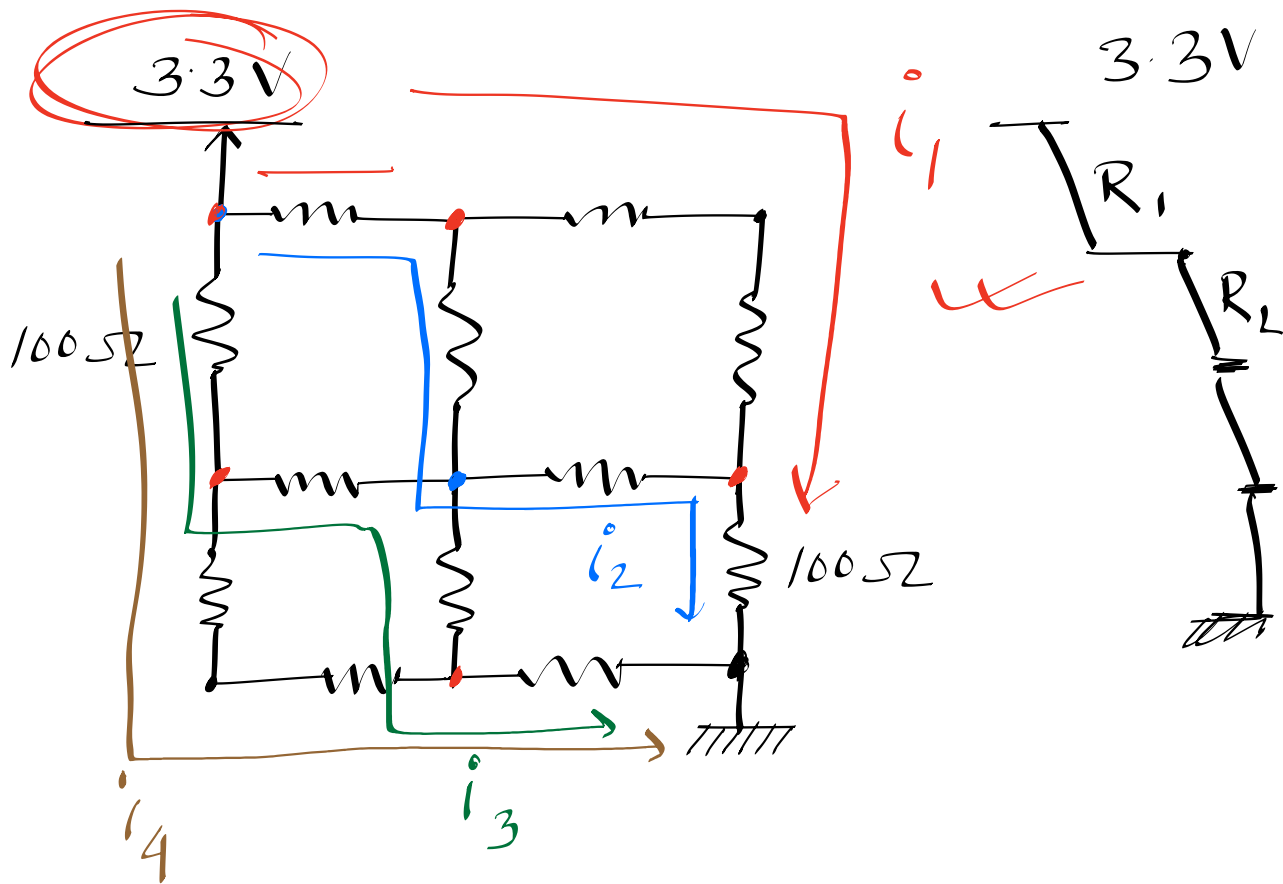


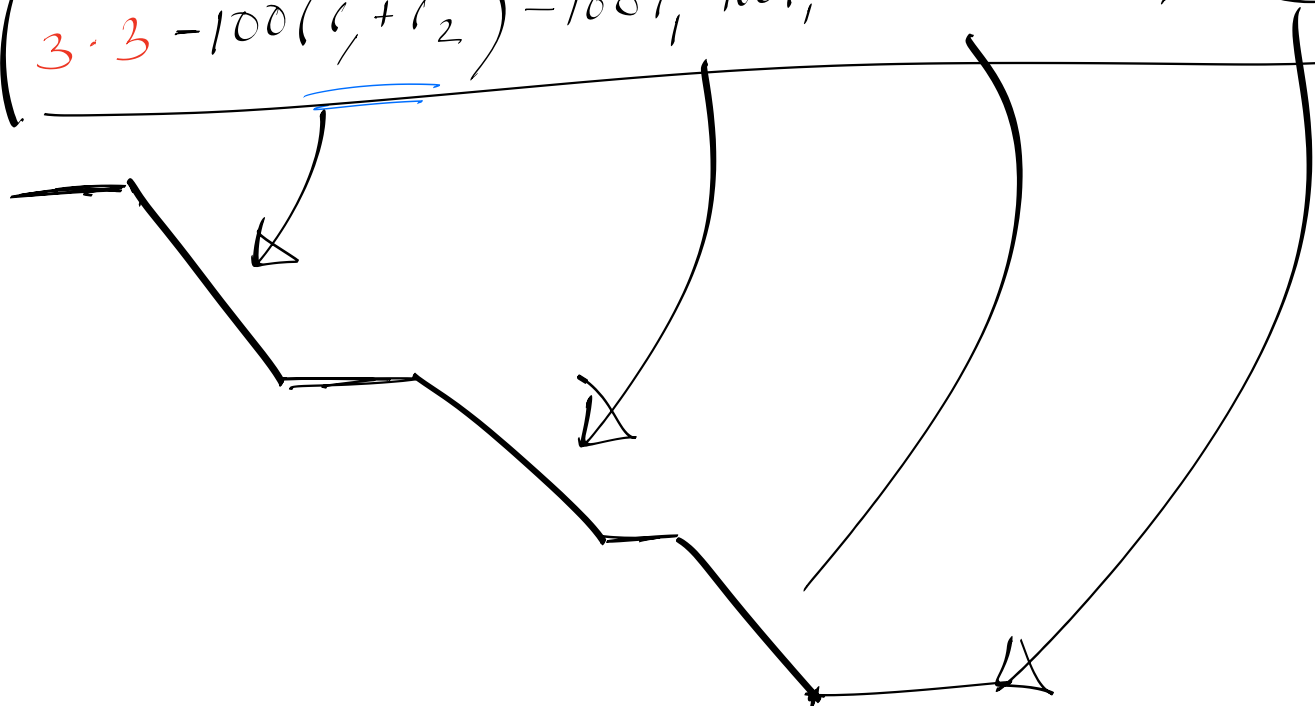
Symmetric Touchpanel

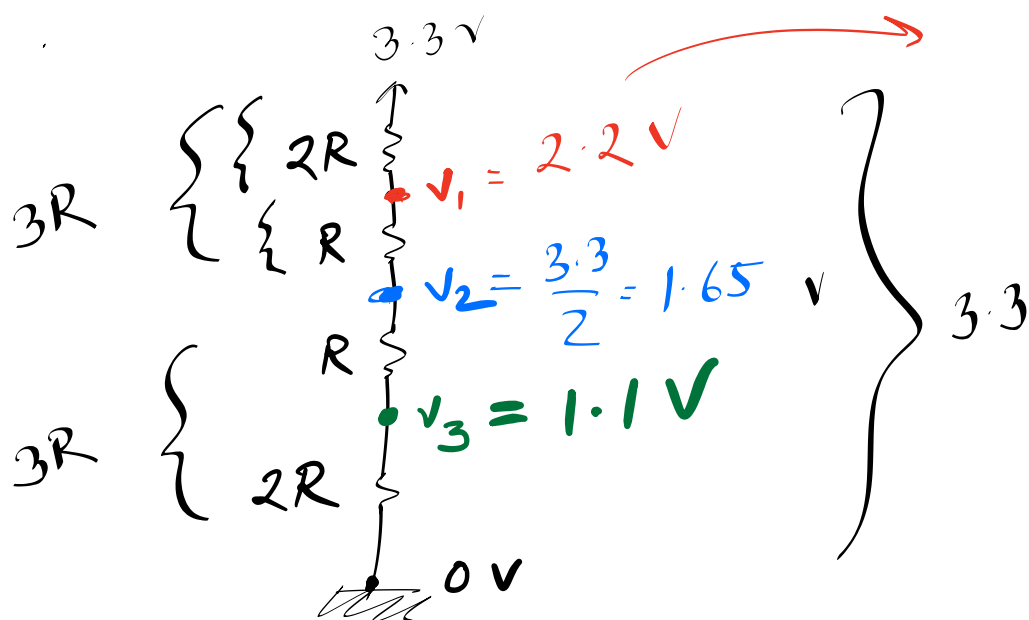
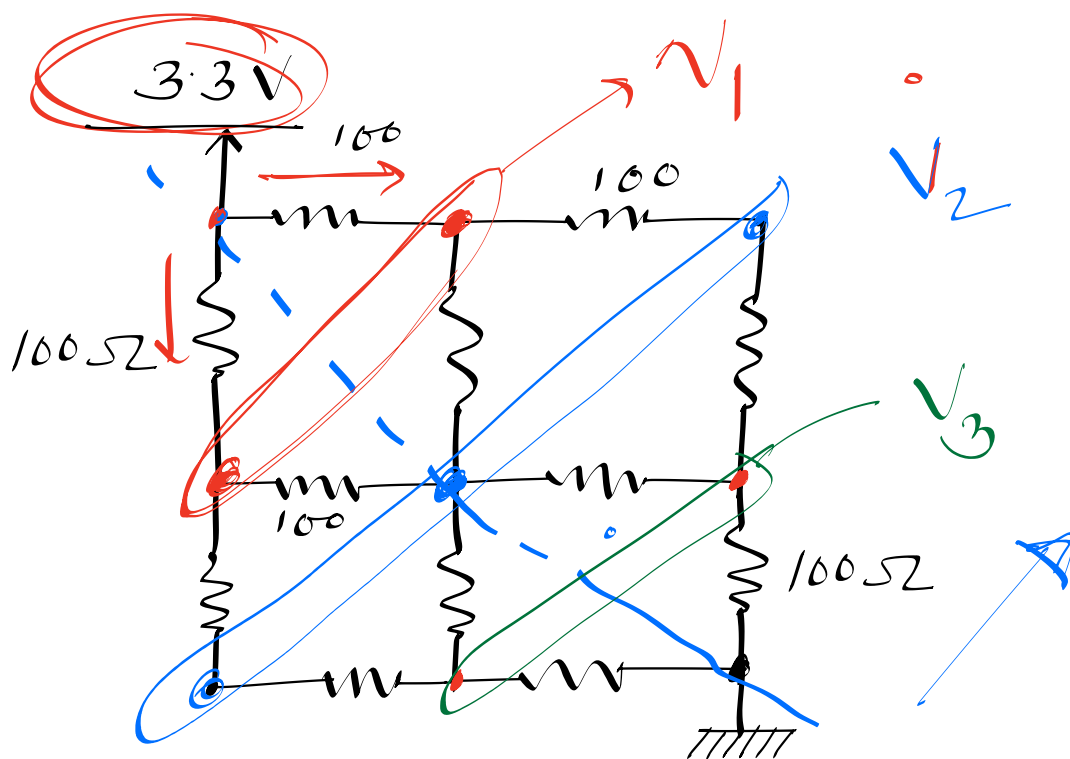


KVL:

KVL:

$\rightarrow 3 \cdot 3 - 100(i_1' + i_2') - 100i_1' - 100i_2' - 100(i_1' + i_2') = 0$

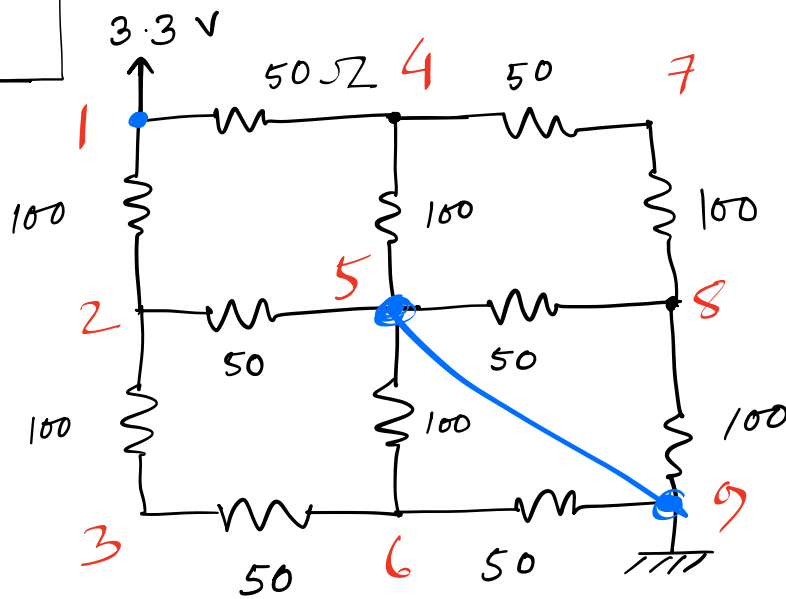




$$V_1 = \frac{R + R + 2R}{R + R + 2R + 2R} \times 3.3$$

$$1:2 = \underbrace{0.55:1.1}_{3.3}$$

Asymmetric Touchpanel

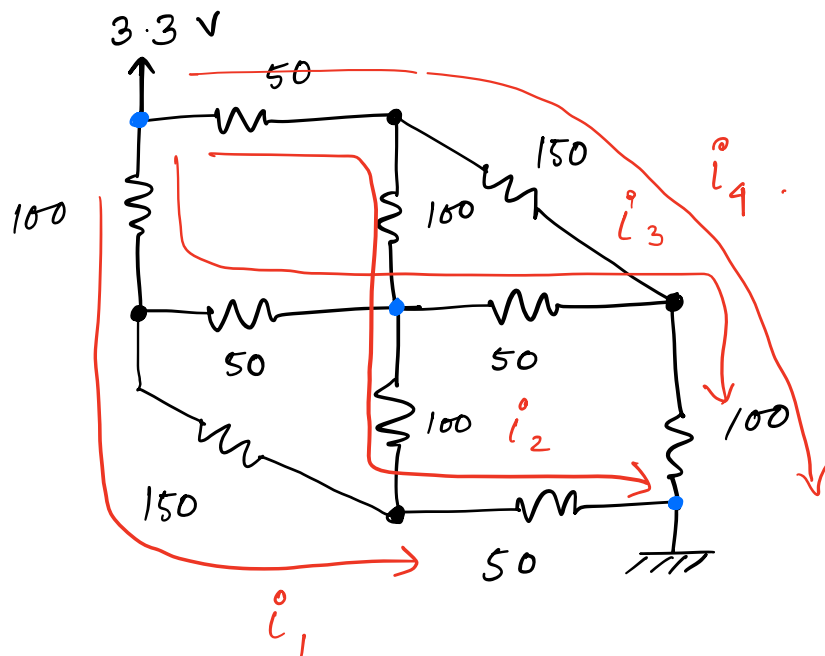


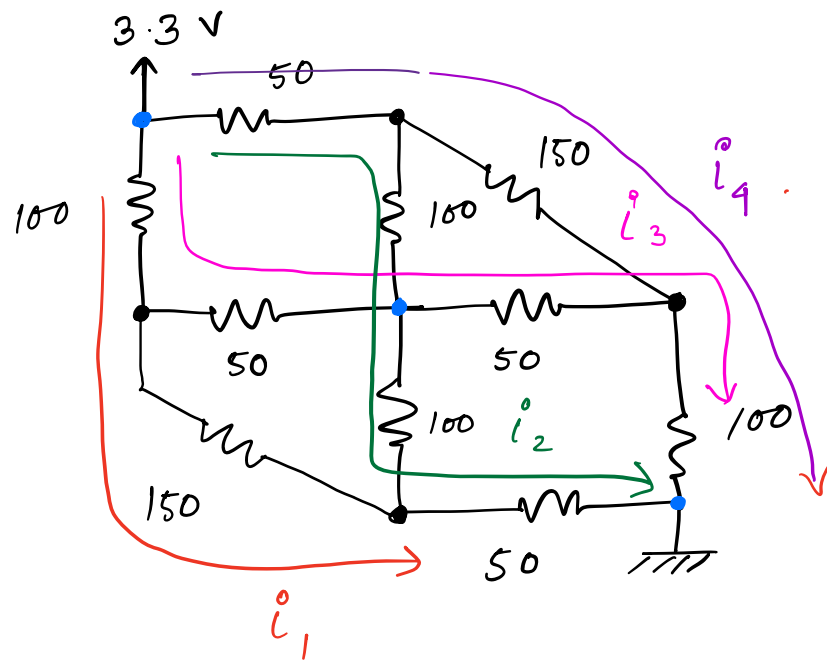
2 bits

0 / 1

$\begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$
 $\begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$

$$4 = 2^2$$





12 bit:

$$v = \frac{s \cdot v}{4095} \times 3.3 \checkmark$$

R.v $\rightarrow$ $\boxed{\frac{v}{3.3} = \frac{s_v}{4095}} \rightarrow 12 \text{ bit}$

$\checkmark$ R.v $\rightarrow$ $\boxed{\frac{v}{5} = \frac{s_v}{1023}} \rightarrow 10 \text{ bit}$

$\boxed{10 \text{ bits}} \rightarrow \left. \begin{array}{c} 0 \quad 000 \\ 10 \quad 000 \\ \text{values} \end{array} \right\} 8 = 2^3$
 $\boxed{0 \rightarrow (2^{10} - 1)}$
 $7 \quad 1 \quad 1 \quad 1$
 $0 \rightarrow 1023$

$$2^{10} \rightarrow 1024$$

$$2^{12} \rightarrow \boxed{4096}$$

$$\boxed{0 \rightarrow 4095}$$